Spanning Tree Protocol (STP)

Spanning Tree Protocol (STP) is a Layer 2 network protocol used in Ethernet networks to prevent loops. In a switched network, loops can accidentally occur when multiple switches are connected with redundant paths. While redundancy is good for reliability, loops can overwhelm the network with broadcast traffic and cause serious performance issue. STP solves this by creating a loop-free logical topology.

STP works by selecting one switch to act as the Root Bridge, which becomes the center of the network's spanning tree. All other switches determine the shortest path to this Root Bridge based on bridge IDs and port costs. This helps organize how data should flow through the network in a predictable way.

After choosing the Root Bridge, STP decides which ports ~~should~~ should forward traffic and which ports should be blocked. Forwarding ports help carry normal traffic, while blocked ports serve as backups in case the main path fails. This ensures that only one active path exists between any two network points, eliminating loops while still keeping backup paths ready.

STP uses a set of states like; Blocking, Listening, Learning and Forwarding to transition ports safely without disrupting the network. These stages help a switch understand the network before it starts sending real traffic, preventing sudden changes from causing loops.